
Intelligence Engineering

Solving Real World Problems At Scale

Prof. Dr.-Ing. Mark Schutera | September 25, 2026 · *drifting stream*

LECTURE III

Data Engineering Fundamentals

This lecture in one slide

III Data Engineering Fundamentals

Data Sources

Data Formats

Data Models

Data Storage Engines and Processing

Modes of Dataflow

Batch Processing Versus Stream Processing

*why this lecture matters, and what
you can do after it*

1

Data Sources

Data Sources

$P(y|x)$

[1]

2

Data Formats

JSON

JSON

Row-Major Versus Column-Major Format

Row-major

Column-major

Text Versus Binary Format

Text

Binary

3

Data Models

Relational Model

Definition: Relational model

$P(y|x)$

[1]

NoSQL

$P(y|x)$

Document model

Graph model

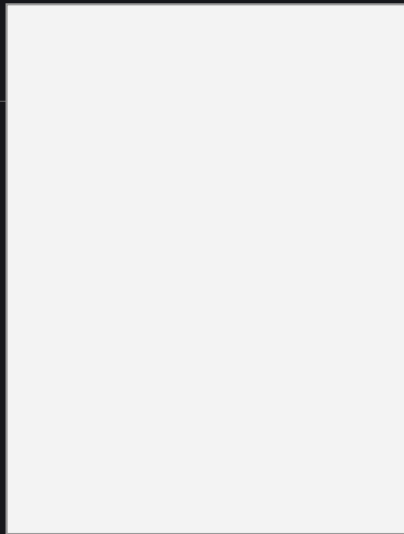
[1]

Document model

$P(y | x)$

[1]

Graph model



$P(y | x)$

[1]

Structured Versus Unstructured Data

Structured data

Unstructured data

4

Data Storage Engines and Processing

Transactional and Analytical Processing

Transactional processing

Analytical processing

ETL: Extract, Transform, and Load

Extract

Transform

Load

5

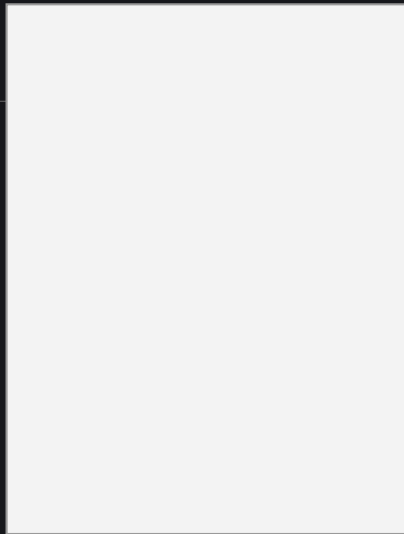
Modes of Dataflow

Data Passing Through Databases

$P(y|x)$

[1]

Data Passing Through Services



$P(y | x)$

[1]

Data Passing Through Real-Time Transport

$P(y|x)$

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[1]

6

Batch Processing Versus Stream Processing

Batch Processing Versus Stream Processing

Batch processing

Stream processing

7

Summary

Summary

$P(y|x)$

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[1]

References

- 1 Chip Huyen. *Designing Machine Learning Systems: An Iterative Process for Production-Ready Applications*. Sebastopol, CA: O'Reilly Media, 2022.